國立臺北大學
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1142_Generative AI Innovative Applications (M6031) - Team 4

ElderEase: A Generative AI-Based Smart Medication Adherence System for Elderly Care-

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Introduction

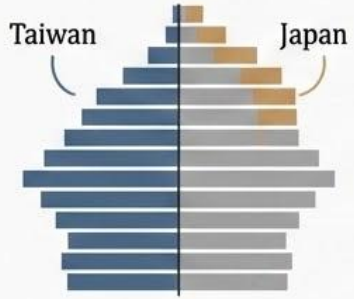
1.1 Research Background

Globally, the proportion of **older adults** is rising. As **aging populations** grow, ensuring proper healthcare management – particularly consistent **medication adherence** – has emerged as one of the most pressing challenges in modern healthcare systems.

Older adults frequently **manage multiple chronic conditions** requiring strict medication schedules. However, age-related cognitive decline, complex multi-drug regimens, and limited day-to-day supervision create significant risks of **missed doses, incorrect intake, and dangerous drug interactions** – all of which contribute to preventable hospitalizations and escalating healthcare expenditures.

The widespread adoption of smartphones and instant messaging platforms like LINE presents a timely opportunity to bridge this gap. Intelligent, **AI-powered systems** can now proactively **support** elderly users in managing their medications .

1.1 Research Background



Rapidly Aging Population: Healthcare & Medication Adherence become critical concerns.

The Medication Challenge

- Multiple medications at specific times
- **Factors:** Memory decline, complex prescriptions, lack of constant supervision
- **Risk:** Forgetting or incorrect dosage

Negative Outcomes & Costs

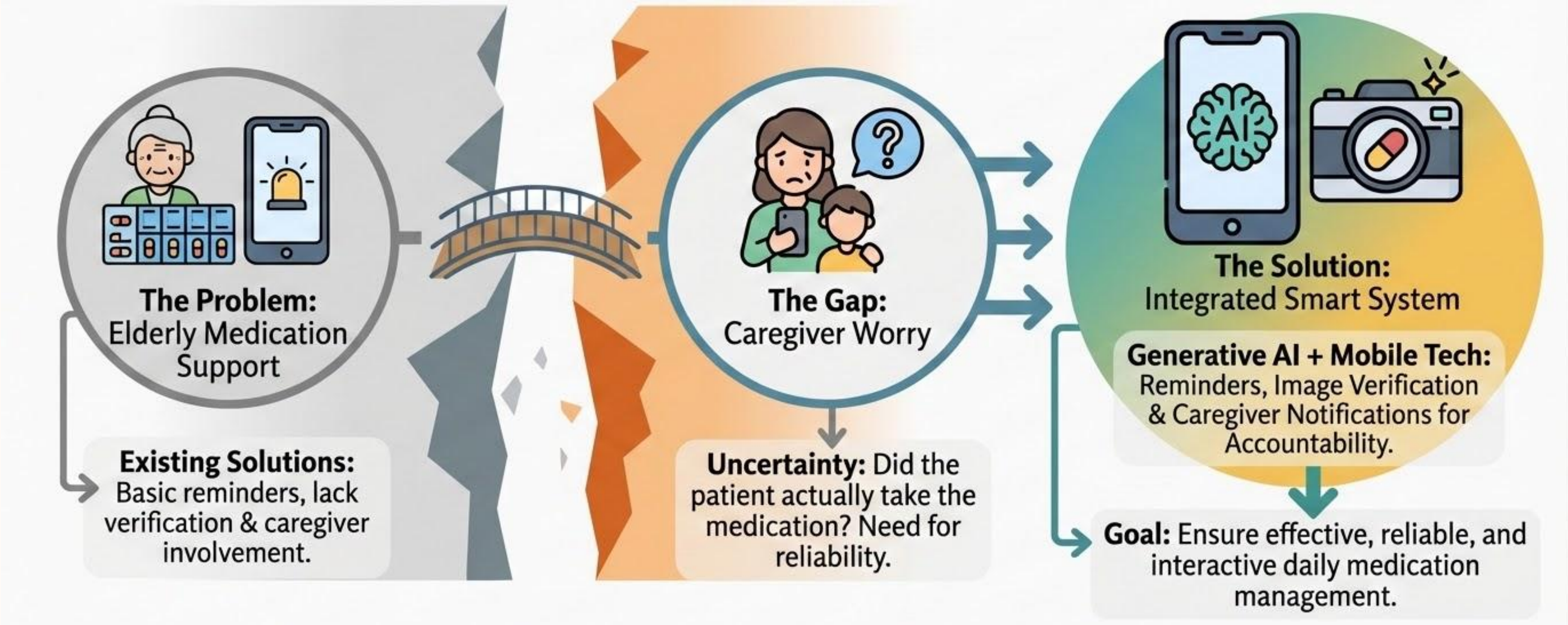
Serious health complications, hospitalizations, and increased healthcare costs.

The Technological Opportunity

Leveraging Mobile Apps & Communication Platforms (e.g., LINE)

Develop smart systems to assist elderly & keep caregivers informed in real-time.

1.2 Research Motivation



1.3 Research Objective

The main objective of this research is to develop a smart medication adherence system that enhances reliability and communication between elderly users and caregivers.

The specific objectives include:

- To **design a mobile application** that provides **timely medication reminders** for elderly users
- To implement a **verification mechanism** where users capture and send images of taken medication
- To integrate with **LINE** for real-time caregiver notifications
- To **improve medication adherence** and reduce health risks associated with missed doses

02

User Scenario

A Day in the Life with ElderEase

"From Reminder to Verification — Closing the Gap in Elderly Medication Care."

User Profile

Mr. Chen, 72

Lives alone, manages 3 medications for hypertension and diabetes. Daughter works full-time.

The Outcome

Peace of mind for family. Consistent daily compliance for Mr. Chen. Feeling supported, not monitored.

08:00 AM — Step 1: Morning Reminder

Notification: "*Grandpa Grandma ! Time for your medication.*"

08:05 AM — Step 2: Verification

Mr. Chen sends a photo of pills. AI confirms correct medication taken.

08:06 AM — Step 3: Caregiver Notification

Automatic LINE alert to daughter: "*Dad has taken medication successfully.*"

01:00 PM — Step 4: Missed Dose Alert

Follow-up reminder and daughter notification if dose is missed.

09:00 PM — Step 5: Evening Summary

Daily adherence report sent to both for complete transparency.

Define our target audience

01. Demographics

- Age: 60-85 years old (primary), 35-55 (caregivers)
- All genders
- Taiwan (urban & suburban); scalable to aging Asian societies
- Retired elderly
- Middle income; reliant on NHI healthcare system

02. Psychographics

- Values family connection and peace of mind
- Cautious about new technology but open if simple
- Trusts LINE as a familiar daily communication tool
- Motivated by health maintenance and independence

03. Behavior

- Takes multiple medications daily at fixed times
- Uses LINE messaging app regularly
- Low engagement with complex health apps
- Relies on family reminders for medication

04. Pain points

- Forgetting medication doses due to memory decline
- No way for caregivers to confirm medication was actually taken
- Existing apps require too many steps or are not senior-friendly

05. Aspirations

- Elderly want to stay independent and healthy at home
- Caregivers want assurance without constant check-in calls

06. Communication

- Primary: App + LINE
- Secondary: Mobile push notifications

07. Customer

- Elderly patient managing chronic conditions
- Healthcare worker or nurse managing multiple elderly patients

08. Customer journey

- Awareness
- Onboarding
- Daily Use
- Monitoring
- Retention
- Expansion

03

Literature Review

2.1 Literature Review

Method	Limitation
Basic Reminder Apps	No verification of actual intake
IoT Sensor Systems	High hardware cost & complexity
Rule-Based Chatbots	Lack personalization
SMS / Phone Reminders	No caregiver feedback loop
Reinforcement Learning	Large data requirement

Observation

Existing solutions provide basic reminders but lack real-time medication intake verification and active caregiver involvement — creating a critical reliability gap.

2.2 Literature Summary Table

Author	Technique / Focus	Limitation
Kumar et al. (2024)	AI Health Management for Elderly	Lacks medication verification
Kapse et al. (2025)	IoT + Deep Learning (MediServe)	High hardware dependency
Patrick & Light (2025)	AI for Medication Compliance	No real-time caregiver alerts
Tana et al. (2024)	AI Integration in Geriatric Care	No intake confirmation loop
Ababtain et al. (2025)	SmartCare Mobile App	Limited AI interaction layer
ElderEase (Proposed)	**GenAI + Image Verification + LINE**	**Explainable & caregiver-connected**

04

Architecture

4.1 SYSTEM ARCHITECTURE

LAYER 1: USER INTERFACE

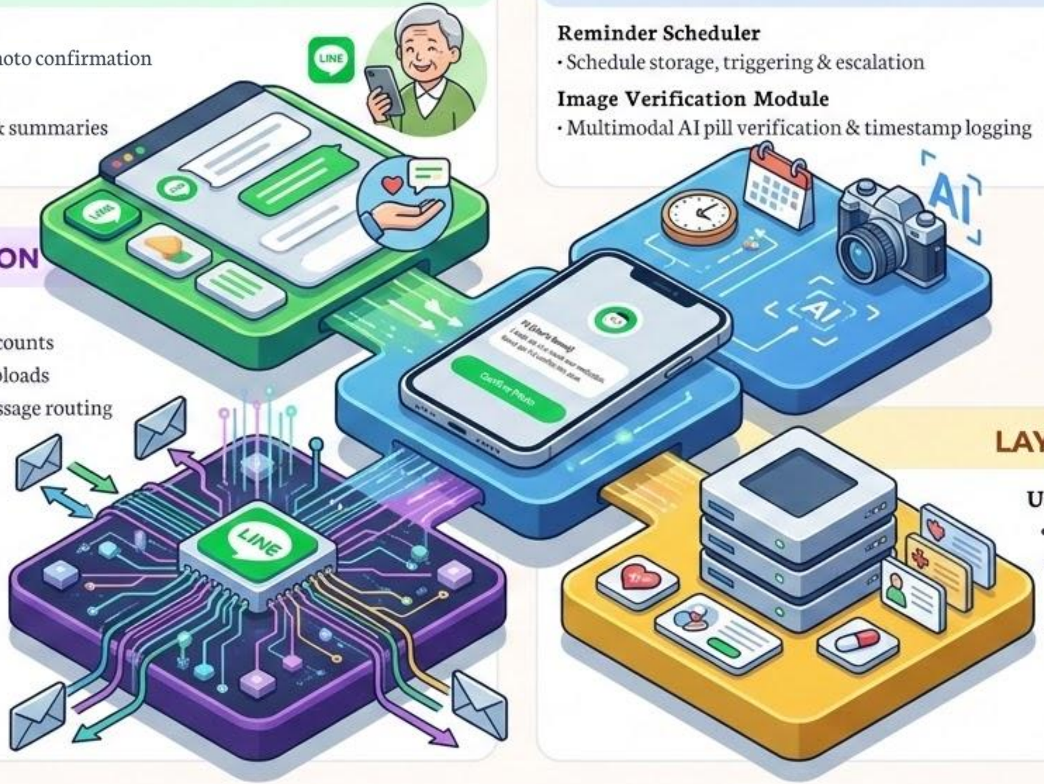
Elderly User (Patient)
- LINE Chatbot+App: AI reminders & photo confirmation

Caregiver / Family Member
- LINE Notifications: Real-time alerts & summaries

LAYER 2: APPLICATION LOGIC

Reminder Scheduler
- Schedule storage, triggering & escalation

Image Verification Module
- Multimodal AI pill verification & timestamp logging

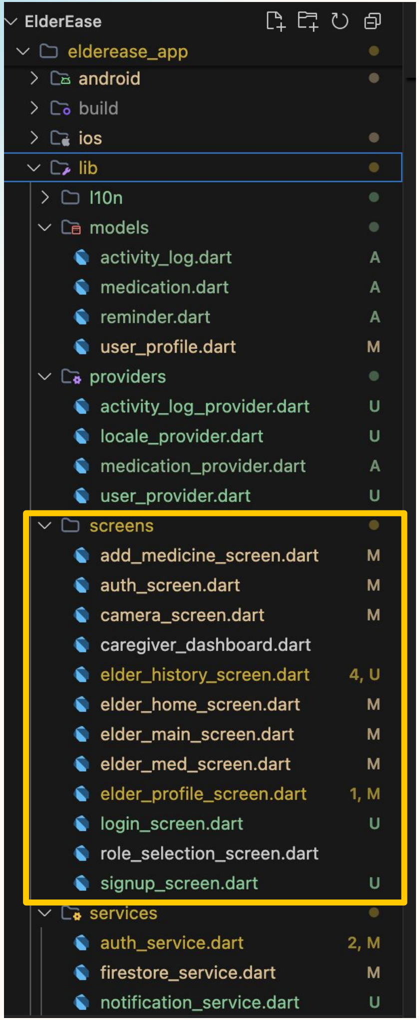


LAYER 3: INTEGRATION

LINE Messaging API
- Connects chatbot to user accounts
- Handles incoming media uploads
- Manages group & direct message routing

LAYER 4: DATA LAYER

User Profile & Medication DB
- Health conditions & preferences
- Visual pill references & interaction warnings
- Caregiver settings & contact info



4.2 Software Architecture

LAYER 1 — Presentation Layer (Flutter UI)

Authentication Flow

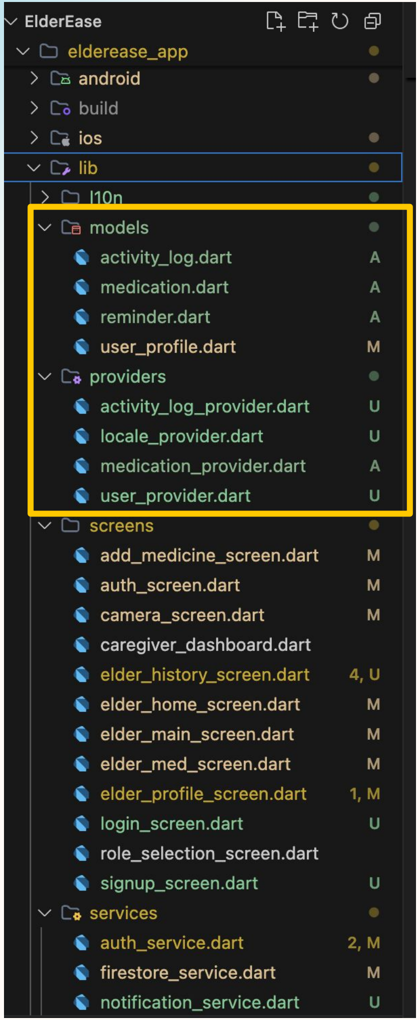
- `login_screen.dart` — Email/password login via Firebase Auth
- `signup_screen.dart` — New user registration
- `role_selection_screen.dart` — Role selection: Elderly or Caregiver

Elderly User Screens

- `elder_main_screen.dart` — Nav shell
- `elder_home_screen.dart` — Dashboard
- `elder_med_screen.dart` — Med list
- `add_medicine_screen.dart` — Med setup
- `camera_screen.dart` — AI photo
- `elder_history_screen.dart` — Intake logs

Caregiver Screens

- `caregiver_dashboard.dart` — Real-time status, logs, & missed dose alerts



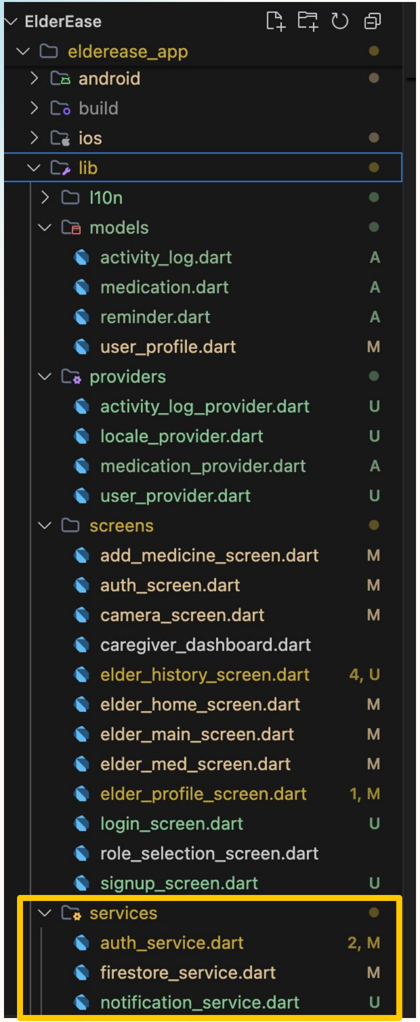
4.3 Software Architecture

LAYER 2 — State Management Layer (Provider)

- `user_provider.dart` — Manages current logged-in user state and role
- `medication_provider.dart` — Handles medication list, CRUD, and schedule
- `activity_log_provider.dart` — Manages real-time activity and adherence log
- `locale_provider.dart` — Manages language and localization (l10n)

LAYER 3 — Data Model Layer

- `user_profile.dart` — Profiles, roles (elder/caregiver), preferences
- `medication.dart` — Data: dosage, frequency, and scheduled times
- `reminder.dart` — Scheduled time, status (confirmed/missed), links
- `activity_log.dart` — Entry: timestamp, action, results, photo ref



4.4 Software Architecture

LAYER 4 – Service Layer

`auth_service.dart`

- Firebase Authentication integration
- Handles sign up, login, logout, and session persistence
- Role assignment stored in Firestore on registration

`firestore_service.dart`

- All Firestore read/write operations
- Stores and retrieves: user profiles, medications, reminders, activity logs
- Real-time listeners for caregiver dashboard live updates

`notification_service.dart`

- Triggers audio: *"Grandma, Grandma, take your medicine!"*
- Sends Firebase push notification at scheduled time
- Missed dose alert to caregiver via LINE (after 5 min)
- Confirmation notification to caregiver via LINE

05

App Demo

Please Scan The QR Code!

SCAN ME

https://github.com/lokendradevv/ElderEase

06

Business Model

6.1 Business Model

KEY PARTNERS
- LINE Taiwan (Platform)
- Hospitals & Clinics (Distribution)

KEY ACTIVITIES
- Train & improve AI models
- Maintain reminder systems
- Monitor data & security

KEY RESOURCES
- LLM & vision AI models
- LINE API infrastructure

VALUE PROPOSITION
- Reminders personalized to routines
- Real-time photo verification of intake
- Instant caregiver notifications via LINE
- Simple, senior-friendly interface
- Reliable daily adherence tracking & reporting

CUSTOMER RELATIONSHIP
- Freemium access
- Daily AI interactions
- Weekly caregiver reports

CHANNEL
- LINE Official Account
- Hospital/clinic referrals

CUSTOMER SEGMENT
- Elderly adults (60+) managing medications
- Adult children & family caregivers
- Elderly care facilities
- Hospitals & healthcare providers

COST STRUCTURE
- AI API usage costs (LLM + vision model calls)
- Cloud hosting and infrastructure maintenance
- Technical team salaries and development costs
- Marketing, partnerships, and user onboarding

REVENUE STREAMS
- Freemium subscription (premium features for families)
- B2B licensing to hospitals, clinics, and care homes
- Pharmacy and health brand partnerships & co-promotions
- Anonymized adherence data insights for research (opt-in)

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6.2 SWOT ANALYSIS

STRENGTHS

- Intelligent, personalized AI reminders
- Image verification of intake
- Built on trusted LINE platform
- Real-time caregiver alerts
- No extra hardware needed
- Highly scalable

WEAKNESSES

- Depends on smartphone & internet
- Elderly may need guidance with photos

OPPORTUNITIES

- Taiwan's rapidly aging population
- Govt seeking digital health tech
- LINE dominance in Taiwan
- Post-COVID trust in AI health

THREATS

- Strict PDPA security regulations
- Tech discomfort among elderly

04

Future Scopes and Conclusions

4.1 Future Scopes for ElderEase

01. Multi-Language Support

Mandarin, Taiwanese Hokkien, and Hakka support to improve accessibility for all literacy levels.

02. AI Drug Detection

Safety layer to flag dangerous drug combinations based on full prescription history.

03. Wearable Integration

Sync with smartwatches to correlate adherence with heart rate, blood pressure, and activity.

04. Disease Management

Comprehensive platform for diet tracking, blood sugar monitoring, and appointment scheduling.

05. Predictive Alerts

ML on historical patterns to predict and prevent missed doses before they happen.

06. NHI System Integration

Collaboration with Taiwan’s NHI for automatic medication schedule imports from prescriptions.

4.2 Final conclusions

01	02	03	04	05
Generative AI significantly **enhances medication** adherence by delivering **intelligent, personalized reminders** beyond basic scheduled alerts	**Image-based verification** closes the critical gap between **sending a reminder and confirming actual** medication intake	**Integrating with LINE** leverages an already trusted platform, **reducing the adoption barrier** for elderly users in Taiwan	**Real-time caregiver** notifications transform passive monitoring into active, connected family participation in **elderly healthcare**	**ElderEase demonstrates** that AI-powered **healthcare tools** can be both clinically effective and accessible to **non-technical users**

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1142_Generative AI Innovative Applications (M6031) - Team 4 Final Group Presentation

Thank you for listening! Q/A

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